General Biology 1

First Semester

Supplementary Materials

Module 9

Cell Transport Mechanisms

Raymart F. Redita
Author

What I know

Directions: Read each situation carefully and choose the best answer.

1. Which type of transport does NOT require energy from the cell?

a. Active transport
b. Endocytosis
c. Passive transport
d. Exocytosis

2. What happens during diffusion?

a. Movement of water across a membrane
b. Movement of molecules from high to low concentration
c. Movement of molecules using energy
d. Movement of large particles into the cell

3. Which process specifically involves the movement of water across a selectively permeable membrane?

a. Diffusion
b. Facilitated diffusion
c. Osmosis
d. Active transport

4. Which of the following requires energy (ATP)?

a. Diffusion
b. Osmosis
c. Facilitated diffusion
d. Active transport

5. Which process allows materials to enter the cell by forming a vesicle?

a. Exocytosis
b. Diffusion
c. Endocytosis
d. Osmosis

What I Need to Know

This supplementary module is designed to help you understand the **Cell Transport Mechanism**. In this lesson, you will explore the difference between passive and active transport the energy use, direction of movement, and mechanisms involved.

Lesson 9 Cell Transport Mechanisms

Cells maintain balance by controlling how substances move in and out through the cell membrane. This occurs through passive transport (diffusion, osmosis, and facilitated diffusion), which does not require energy, and active transport (pumps, endocytosis, and exocytosis), which uses energy to move materials. These transport mechanisms are essential for maintaining homeostasis and supporting cell survival.

After going through this supplementary module, you are expected to:

1. identify and describe the different types of passive transport (diffusion, osmosis, and facilitated diffusion) and explain how substances move across the cell membrane without energy;
2. explain the processes of active transport (membrane pumps, endocytosis, and exocytosis), including how energy is used to move substances against the concentration gradient; and
3. compare and differentiate passive and active transport in terms of energy use, direction of movement, and mechanisms involved.

What's In

Activity 1: "No Energy Needed: The Passive Transport Experience"

Directions: Match and move: Identify and describe diffusion, osmosis, and facilitated diffusion by matching each process with its correct description and real-life example.

Process	Description	Example
Diffusion		
Osmosis		
Facilitated Diffusion		

Movement of water across a selectively permeable membrane	Perfume spreading in a room
Movement of molecules through protein channels	Glucose entering the cell through protein channels
Movement of molecules from high to low concentration	Plant cells swelling or shrinking

Passive transport is the movement of substances across the cell membrane without the use of energy (ATP). In all forms of passive transport, particles move from an area of **higher concentration to lower concentration**, helping the cell maintain balance or **homeostasis**. There are three main types: **diffusion**, **osmosis**, and **facilitated diffusion**.

Diffusion is the simplest form of passive transport, where small molecules like oxygen and carbon dioxide move directly across the cell membrane. These molecules spread out until they are evenly distributed, reaching a state of equilibrium.

Osmosis is the diffusion of water molecules across a selectively permeable membrane. Water moves from an area of higher water concentration (lower solute concentration) to an area of lower water concentration (higher solute concentration). This process is important for maintaining the proper water balance in cells. For example, plant cells can swell or shrink depending on the surrounding solution.

Facilitated diffusion involves the movement of larger or charged molecules, such as glucose and ions, across the membrane with the help of protein channels or carrier proteins. Although it requires assistance from these proteins, it still does not use energy and follows the concentration gradient.

Overall, passive transport allows essential substances to enter and exit the cell efficiently without energy, ensuring that cells function properly and remain in a stable internal environment.

What's New

Activity 2: "Against the Flow: The Active Transport Mission"

Directions: Analyze the scenario given and identify the types of transport. Choose for the card below.

Membrane Pumps	**Endocytosis**	**Exocytosis**

Scenario	**Type of Transport**	**Why is Energy is Needed?**
A cell uses energy to move ions into an area of high concentration		
A cell engulfs bacteria		

A cell releases waste material		

Guide Questions:

1. How is active transport different from passive transport in terms of energy use? _____
   _____
2. Why do cells need to move substances against the concentration gradient? _____
   _____
3. What might happen if a cell cannot perform active transport?
   _____
   _____

What's More

Activity 3: "Transport Battle Cards"

Directions: Using your knowledge of cell transport, fill in the table by comparing passive transport and active transport.

Feature	Passive Transport	Active Transport
Energy Use		
Direction of Movement		
Mechanisms		

Guide Questions:

1. How does energy use differ between passive transport and active transport? _____
_____
_____

2. In what direction do substances move in passive transport compared to active transport? _____
_____
_____

3. What mechanisms or processes are involved in each type of transport?
_____
_____
_____

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What I've Learned

Directions: Encircle the letter of the correct answer.

1. A cell membrane protein stops functioning, preventing glucose from entering the cell without energy. Which process is directly affected?

a. Diffusion
b. Osmosis
c. Facilitated diffusion
d. Exocytosis

2. A cell needs to move ions into an area where their concentration is already high. Which transport mechanism will the cell most likely use?

a. Diffusion
b. Osmosis
c. Facilitated diffusion
d. Active transport

3. Why would a cell use active transport instead of passive transport?

a. To save energy
b. To move substances quickly
c. To move substances against the concentration gradient
d. To allow water to enter the cell

4. A cell is placed in a solution where molecules are evenly distributed inside and outside. What will most likely happen to diffusion?

a. It will increase rapidly
b. It will stop because equilibrium is reached
c. It will require energy to continue
d. It will turn into active transport

5. Which situation best demonstrates the combined importance of passive and active transport?

a. A cell only uses diffusion for all processes
b. A cell uses active transport for all movement
c. A cell uses passive transport for balance and active transport for specific needs
d. A cell stops transporting materials